ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES

DISTRIBUTED SYSTEMS (CCS 4312) DECEMBER 2024 FINAL EXAM

INSTRUCTIONS

(1) Time allowed is **Three (3) hours**.
(2) There are **Six (6)** questions in this paper and you are required to answer any **Four (4).**
(3) Each question carries **Twenty Five (25)** marks.
(4) Calculators are not allowed.
(5) The total marks for this Exam is **100**

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Question One

i. The distributed shared memory abstraction is implemented by using the services of the underlying message passing communication system. In principle, the performance of applications that the use DSM is expected to be worse than if they use message passing directly.
   a) Explain why some distributed operating system designers support the DSM abstraction in their systems. **[5 Marks]**
   b) Are there any applications that can have better performance in a system with DSM facility than in a system that has only message passing facility? If yes, give the types of such applications. If no, explain why. **[5 Marks]**

ii. Discuss the relative advantages and disadvantages of using large block size and small block size in the design of a block based DSM system. **[5 Marks]**

iii. Explain why most DSM system designers prefer to use the typical page size used in a conventional virtual memory implementation as the block size of the DSM system. **[5 Marks]**

iv. It is often said that the structure of the shared memory space and the granularity of data sharing in a DSM system are closely related. Explain why? **[5 Marks]**

Question Two

i. Elaborate the operation of Lightweight RPC. **[10 Marks]**

ii. List and explain the special RPC models. **[5 Marks]**

iii. The caller process of an RPC must wait for a reply from the callee process after making a call. With the use of clear examples, explain how this process is achieved. **[5 Marks]**

iv. Discuss how optimization of RPC is done to enhance the performance. **[5 Marks]**

Question Three

i. What was the primary motivation for development of RPC? **[5 Marks]**

ii. Explain the main difference between RPC model and an ordinary procedure call model. **[5 Marks]**

iii. Use a well labeled diagram to discuss the request / reply / acknowledge reply (RRA) protocol model as used in RPC. **[15 Marks]**

iv. What are the issues in developing a transparent RPC mechanism? **[5 Marks]**

v. State the functionality and purpose stubs. **[5 Marks]**

Question Four

i. In RPC based applications, two important issues that need to be considered for every management are server implementation and server creation. Distinguish between the two concepts. **[12 Marks]**

ii. Explain in detail the architecture and principles of Monolithic kernel and Micro kernel. **[4 Marks Each]**

iii. List the five (5) elements of program that are involved in the implementation of an RPC mechanism. **[5 Marks]**

Question Five

i. With use of clear examples, explain how location, relocation and migration transparencies differ from each other. **[15 Marks]**

ii. Briefly discuss the security aspects of a Distributed Systems. **[5 Marks]**

iii. Discuss the issues involved in building a DSM system on a network of heterogeneous machines? **[5 Marks]**

Question Six

i. Differentiate between **time-sharing**, **parallel processing**, **network** and **distributes operating systems**. **[2 Marks Each]**

ii. In what respect are distributed computing systems better than parallel processing systems? **[5 Marks]**

iii. Discuss the main guiding principles that a distributed operating system designer must keep in mind for good performance of the system? **[7 Marks]**

iv. What are the major issues of designing a Distributed OS? **[5 Marks]**